



ExU-Trans: a self-explanatory transformer with U-Net based hybrid model for brain tumor segmentation using MR imaging

G. M. Sasikala¹ · K. Anand²

Received: 13 November 2025 / Accepted: 20 February 2026 / Published online: 16 March 2026
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Abstract

Brain tumor segmentation from MRI images is a critical task for accurate diagnosis and treatment planning. While traditional segmentation approaches leverage deep learning models, such as U-Net and its variants, they often struggle with balancing precision and interpretability, especially in handling noisy and complex MRI data. Earlier works have demonstrated remarkable performance improvements through hybrid models, yet limitations persist in effectively capturing both global and local contexts, as well as ensuring model transparency for clinical application. These shortcomings highlight the need for frameworks that not only excel in segmentation accuracy but also provide explainability for reliable medical decision-making. To address these challenges, we propose a novel Explainable U-Net Transformer (ExU-Trans) framework for brain tumor segmentation in MRI images. ExU-Trans integrates the interpretability strengths of Vision Transformers with the spatial precision of U-Net in a dual-encoder structure. The Vision Transformer encoder employs Self-Explanatory Multi-Head Attention (SE-MHA) to generate noise-resistant attention maps, while a modified U-Net encoder focuses on extracting multiscale spatial features critical for tumor delineation. To enhance interpretability, a Discriminative Attribute Explainer (DAE) is embedded within the transformer encoder, pinpointing tumor-specific attributes at a pixel level. Furthermore, a Contextual Self-Attention mechanism enriches feature representations by capturing both global and local contexts, refining tumor detection. The outputs from both encoders are combined using a bivariate fusion module that incorporates spatial and channel attention, producing a unified feature representation. Finally, an attribute-guided loss function enables self-supervised learning, resulting in highly accurate and inherently interpretable segmentation outcomes. Extensive evaluations demonstrate the superiority of ExU-Trans in terms of segmentation accuracy and explainability, offering a reliable tool for clinical brain tumor diagnosis and treatment planning.

Keywords Brain tumor segmentation · MRI imaging · Explainable AI · U-Net transformer · Self-explanatory attention · Multiscale feature extraction

Introduction

Brain tumors are among the most life-threatening and complex conditions affecting the central nervous system (CNS),

significantly impacting a patient's quality of life, cognitive function, and survival rate. The term "brain tumor" encompasses a wide range of abnormalities, from benign growths to highly malignant and aggressive cancers. These tumors vary widely in terms of their growth patterns, biological aggressiveness, and locations within the brain, which presents unique diagnostic and treatment challenges [1–3]. The heterogeneity of brain tumors ranging from gliomas to metastatic tumors requires careful and precise diagnosis to ensure that patients receive optimal and personalized treatment. Early and accurate detection of brain tumors is crucial, as it can significantly improve prognosis by allowing timely interventions. Treatment strategies, such as surgery, radiotherapy, and chemotherapy, heavily rely on tumor size, location, and type, all of which are best evaluated through

✉ G. M. Sasikala
sasikalagm@gmail.com

K. Anand
anand.k@rajalakshmi.edu.in

¹ Department of Artificial Intelligence and Machine Learning, Saveetha Engineering College, Thandalam, Chennai 602105, India

² Department of Computer Science and Engineering, Rajalakshmi Engineering College, Thandalam, Chennai 602105, India

medical imaging modalities [4, 5]. Magnetic Resonance Imaging (MRI) plays a pivotal role in the diagnosis and management of brain tumors due to its non-invasive nature, high spatial resolution, and superior soft tissue contrast. MRI provides high-quality images that capture both the anatomy and the tissue composition of the brain, allowing clinicians to detect abnormal growths, assess their boundaries, and plan interventions. One of the major advantages of MRI is its ability to provide multiple imaging sequences, such as T1-weighted, T2-weighted, and post-contrast images, each revealing different tissue characteristics [6–8]. These sequences are crucial in distinguishing tumors from surrounding healthy tissues, providing insight into tumor type (e.g., cystic, solid) and its vascularity (e.g., enhanced with contrast), all of which influence treatment decisions.

However, while MRI is a powerful tool, interpreting the scans for brain tumor detection and segmentation remains a significant challenge due to the complex and heterogeneous nature of brain tumors, which often blend with surrounding tissues and structures. Tumors may present as subtle changes in intensity, shape, and size, which makes accurate segmentation of tumor regions a complex task [9–11]. Radiologists rely on their expertise to make manual annotations, but human error, fatigue, and the inherent complexity of MRI data often result in inaccurate or inconsistent segmentation [12]. To address these challenges, there has been a growing interest in the development of computational tools to assist radiologists in segmenting brain tumors in MRI images accurately and efficiently. Automated brain tumor segmentation has become a prominent research area, driven by the need to reduce the manual burden on clinicians, minimize diagnostic errors, and provide objective, reproducible assessments [13–15]. Early methods for automatic segmentation relied heavily on handcrafted features, such as texture, shape, and intensity-based descriptors. These features were extracted manually from MRI scans and used to train machine learning models, including classical algorithms like support vector machines (SVMs) and random forests. While these traditional methods showed some promise, they faced significant limitations in their ability to generalize across diverse datasets, especially given the vast variability in tumor types, sizes, and locations [16, 17]. The reliance on manual feature engineering also made the models inflexible and labor-intensive, limiting their scalability in clinical practice.

The advent of deep learning, particularly convolutional neural networks (CNNs), marked a significant breakthrough in medical image segmentation. CNNs have proven to be particularly effective in learning hierarchical spatial features directly from raw data, without the need for manual feature extraction. CNN-based models, such as U-Net, have been widely adopted in medical image analysis due to their ability to perform pixel-level segmentation with high

accuracy [18–20]. U-Net, in particular, is known for its encoder-decoder architecture, which allows it to capture both fine-grained local details and context from larger regions. This ability to localize tumors of varying shapes and sizes has made U-Net and its variants one of the most widely used architectures for brain tumor segmentation. Despite their success, CNN-based models still struggle with capturing long-range dependencies, which are essential for understanding the global context of tumors and their relation to surrounding brain structures. Brain tumors, particularly those located near complex anatomical features, often require the model to consider information from distant regions in the brain to accurately delineate tumor boundaries. While CNNs are excellent at extracting local features, their inability to model long-range relationships limits their capacity to capture the full complexity of brain tumor segmentation. In response to these limitations, transformers, originally designed for natural language processing, have recently gained attention in computer vision tasks, including medical image segmentation. Vision Transformers (ViTs) leverage self-attention mechanisms to model global relationships between different regions of an image, enabling them to capture both local and global features [21–23]. This characteristic makes transformers highly suited for tasks like brain tumor segmentation, where understanding the global context is crucial for accurate tumor localization and boundary delineation. ViTs have demonstrated impressive results in various medical image segmentation tasks, including brain tumor detection. They are particularly beneficial in situations where capturing long-range dependencies and contextual information is critical. However, despite their potential, Vision Transformers in medical image segmentation still face challenges related to computational cost, susceptibility to noise, and difficulty in distinguishing subtle tumor boundaries, which can affect their robustness in clinical applications.

Furthermore, many current transformer-based segmentation models lack interpretability, which is a significant concern in medical applications. In clinical settings, it is essential for models to not only perform well in terms of accuracy but also to provide transparency in their decision-making processes. Interpretability allows clinicians to understand how a model arrived at a particular decision, building trust and confidence in the AI system. While some efforts have been made to make transformer models more explainable, many existing models still provide limited insights into their internal workings, making them less suitable for use in critical healthcare applications where explainability is paramount. Another limitation of existing models is their tendency to overlook crucial attribute-level information that could be vital in distinguishing tumor regions from healthy tissues. Attributes such as texture, shape, and intensity, although implicitly captured by deep learning models, often need to be explicitly emphasized to improve the model's

ability to segment complex tumor structures accurately. For example, subtle differences in tumor texture or irregularities in shape can be key indicators of tumor boundaries but may be overlooked by models that focus solely on pixel-level segmentation. To address these limitations, we propose a novel framework called Explainable U-Net Transformer (ExU-Trans), which combines the strengths of both U-Net and Vision Transformers for brain tumor segmentation in MRI images. Our approach introduces a hybrid architecture that leverages the spatial precision of U-Net with the global context modeling capability of Vision Transformers, along with a focus on explainability. ExU-Trans achieves robust tumor segmentation with improved accuracy and model transparency, making it a valuable tool for both clinicians and researchers in the field of brain tumor diagnosis and treatment.

Research contribution

- **Dual-Encoder Architecture with Explainability :** The proposed Explainable U-Net Transformer (ExU-Trans) framework introduces a novel dual-encoder architecture that integrates a Vision Transformer with a U-Net encoder. The Vision Transformer encoder employs Self-Explanatory Multi-Head Attention (SE-MHA), generating noise-resistant attention maps, while the U-Net encoder ensures precise spatial feature extraction and multiscale localization. This hybrid design enables accurate segmentation of brain tumors by combining global contextual understanding with fine-grained spatial resolution.
- **Discriminative Attribute Explainer (DAE):** DAE is embedded within the Vision Transformer encoder, making the model inherently interpretable by identifying tumor-specific attributes at a pixel level. This innovation enhances transparency and reliability, crucial for clinical applications, by providing detailed insights into the model's decision-making process while highlighting the most relevant tumor features.
- **Bivariate Fusion Module with Attribute-Guided Loss:** The framework employs a Bivariate Fusion Module that combines features extracted from the two encoders through spatial and channel attention mechanisms. This fusion ensures a cohesive representation, amplifying critical features for segmentation. Additionally, an Attribute-Guided Loss Function is introduced to enable self-supervised learning, improving boundary delineation and alignment with tumor regions while enhancing segmentation precision and robustness.

Paper organization

The research paper is structured into five key sections to comprehensively present the proposed ExU-Trans model for brain tumor segmentation. "[Introduction](#)" section (Introduction) highlights the challenges in brain tumor segmentation from MRI images, "[Literature survey](#)" section (Literature Survey) reviews existing deep learning and hybrid models, discussing their limitations in balancing precision, interpretability, and handling noisy, complex data. "[ExU-Trans framework](#)" section (Proposed Methodology) details the ExU-Trans architecture, including the dual-encoder design combining overing its architecture, components, and innovations, ensuring clarity and reproducibility. "[Experimental results](#)" section (Experimental Results) presents a thorough evaluation of the model's performance on standard datasets, comparing it with state-of-the-art methods to showcase its superiority in accuracy and explainability. Finally, "[Conclusion and future work](#)" section (Conclusion) summarizes the contributions of the study, its implications for clinical applications, and potential directions for future research.

Literature survey

In this section, we have articulated the state-of-art in tumor segmentation for better understanding. Furthermore, this section is divided into three sub-section which are articulated as below,

Allah et al. [24] presents nnU-Net architecture, a modified U-Net variation designed for tumor segmentation, is introduced in this study. The nnU-Net makes use of a cutting-edge network architecture with tiered skip connections and a combative learning approach. The outcomes demonstrate outstanding performance in terms of segmentation robustness and accuracy. Additionally, the authors offer an open-source solution that is simple to use, opening it up to the larger research community. Kumar [25] develops Trans-BTS, or Multimodal based tumor segmentation utilizing Transformer, is a novel method that applies the Transformer design, generally employed in NLP, to the segmentation of tumors. The Transformer's applicability to multimodal MRI data is expanded in this article. The method produces precise segmentations by efficiently capturing long-range relationships between picture pixels and modalities. By illustrating the potential of Transformer-based approaches in medical image processing, it opens a new route for the discipline. Ottom et al. [26] is presented an attention mechanism in the research report to enhance the meticulousness of tumor segmentation in multimodal MRI images. The segmentation

outcomes are improved by the attention mechanism's ability to assist the model concentrate on important elements while ignoring unimportant ones. It underlines the importance of attention mechanisms in medical picture segmentation and yields encouraging findings. Neelima et al. [27] developed of the standard U-Net design, U-Net++ includes dense skip connections and more feature levels. This study investigates the use of U-Net++ for segmenting tumors. The findings show that U-Net++ can accomplish cutting-edge performance while being computationally effective. The significance of architectural advancements in deep learning for medical image processing is highlighted by this work. Guan et al. [28] The foremost goal of TBraTS is to guarantee the accuracy and dependability of tumor segmentations. It introduces a brand-new quality control system and a validation framework to rate the segmentation models' precision. As it covers the crucial topic of model trustworthiness, which is particularly significant in medical applications, this study is crucial for the field.

Walsh et al. [29] presents a novel strategy by emphasizing modality-pairing learning, in which the model learns the connections between several MRI modalities for tumor segmentation. This approach achieves outstanding segmentation accuracy by utilizing cross-modal data. The idea of modality pairing illustrates how multimodal data may be used to enhance segmentation results, which is especially useful in clinical contexts. Vankdothu and Hameed et al. [30] introduce an effective 3D residual neural network designed specifically for tumor segmentation tasks is by ERV-Net. The architecture's efficiency and capacity for 3D data handling are a big help in precisely defining tumor boundaries. The significance of improving neural network topologies for real-time or resource-constrained applications is highlighted in this research. Nizamani et al. [31] develops a technique called Cross-Modality Deep Feature Learning focuses learning robust features from many MRI modalities. The model shows enhanced segmentation outcomes by explicitly addressing cross-modality information fusion. The method emphasizes how feature extraction methods can improve comprehension of intricate medical images. Meenakshidevi et al. [32] introduce SwinBTS is an inventive way to segment 3D multimodal tumors using the Swin Transformer architecture. It improves segmentation accuracy by adding a transformer-based technique that incorporates both local and global context. This study illustrates how transformer models may handle data from multiple imaging modalities in medical imaging. Kollem et al. [33] presents Multiple tumor segmentation tasks are carried out simultaneously by the Swia deep multi-task learning framework. This method achieves good segmentation accuracy and generalization across several tumor categories by concurrently optimizing for distinct tumor kinds. This multi-task strategy represents a substantial advancement towards more thorough tumor characterization.

Tabatabaei et al. [34] presents a latent correlation representation learning technique to address the problematic of missing MRI modalities. The program effectively fills in for missing data by learning the relationships between the available modalities, producing reliable and accurate tumor segmentations. This method is very useful in clinical situations where there are gaps in the imaging data. Ting and Liu [35] introduce RFNet in terms of region-aware fusion network that focuses on inadequate multi-modal data. To efficiently integrate information from the numerous modalities, the model makes use of region-specific fusion techniques. RFNet is a useful tool for real-world tumor segmentation tasks because of its capacity to manage partial data. Zhang et al. [36] expand the nnU-Net architecture's segmentation capabilities for tumors. The adjustments improve the network's performance, illuminating the significance of architectural advancements in producing cutting-edge outcomes. Wider adoption in the research community is facilitated by the open-source implementation presented in the study. Huang et al. [37] introduce ME-Net a multi-encoder net framework made specifically for segmenting tumors. The model captures a variety of feature representations by using numerous encoders, which improves segmentation accuracy. The strategy highlights the importance of feature extraction methods in intricate medical picture analysis. Lan et al. [38] presents Znet which unveils a DL method designed specifically for 2D MRI tumor segmentation. Despite operating in a 2D domain, the model performs well, suggesting the possibility of effective and precise 2D segmentation techniques. The adoption of this method may be especially beneficial in scenarios where 3D data gathering is constrained.

Liang et al. [39] improve tumor segmentation by integrating K-means clustering, DL, and synthetic data augmentation. The model improves its capacity to generalize across various tumor types and sizes by utilizing clustering for initial segmentation and enhancing the dataset with synthetic data. This technique offers a workable solution to the problematic of data scarcity in medical imaging. Rao et al. [40] examine deep learning methods to extract useful features from brain MRI images with a focus on feature representation. Segmentation is then performed using the extracted characteristics. This method emphasizes the value of feature engineering in medical image analysis because it has the potential to identify tiny features of tumors that could be invisible to the unaided eye. Lin et al. [41] focus on features at various sizes, the Scale Attention Network (SAN) described in this research improves the segmentation process. This makes it possible to better define tumor boundaries and localize tumors. The segmentation of tumors, where tumors might change in size and shape, benefits greatly from the SAN's ability to handle scale fluctuations. Zhu et al. [42] introduced modified U-Net architecture by DR-Unet104 for segmenting multimodal MRI tumors. The architecture's versatility and sturdiness are

demonstrated by how well it can adapt to many modes. This study highlights the significance of utilizing multimodal data to improve segmentation accuracy. ZongRen et al. [43] to overcome the difficulties of segmenting 3D medical images, a 3D U-Net variation with a multipath residual attention block is suggested in this study. The network's ability to concentrate on pertinent locations is enhanced by the attention mechanism, which aids in more precise tumor identification. This strategy is essential for applications where precise segmentation depends on 3D data.

Mehmood et al. [8] introduces a DL system for segmenting tumors using MRI and artificially created CT images. Synthetic data production can assist close the gap among imaging modalities, and the integration of several imaging modalities shows promise. The accuracy and realism of the synthetic data, however, should be discussed in the paper along with the consequences for the adaptability of the model. Farhan et al. [12] explores Tumor segmentation using image fusion and interpolation. These methods might increase the accuracy of segmentation and the quality of the image. A full analysis of the advantages and disadvantages of various techniques, as well as their computing cost and applicability in clinical situations, should be included in the study. Khaliq et al. [20] propose Segmenting MRI and CT tumor images with an enhanced Gabor wavelet transform and uneven K-means clustering algorithm. Although the use of Gabor wavelets and rough clustering is intriguing, a thorough comparison to present state-of-the-art methods should go along with it. Hassain et al. Deep neural networks have been utilized to perform automatic tumor segmentation across various medical imaging modalities. In healthcare, multimodal imaging is essential, and deep learning has tremendous potential in this area. In addition to considering the model's resilience and adaptability across various patient data sets, the study should offer insights into the network's comprehension. Rani et al. A sophisticated tumor identification method has been developed that integrates a wireframe model with 3D AlexNet to analyze combined CT and MRI scans. The use of a 3D neural network in combination with the combining of wireframe models makes sense given the three-dimensional nature of medical imaging. However, the research ought to go into detail about how well the wireframe model depicts tumors and how useful the method is in actual clinical settings.

A cutting-edge approach integrates the quantum matched-filter methodology with deep spiking neural networks to segment tumor regions in MRI and PET images. The use of quantum computing techniques to provide improved filtering for analysis of images is an exciting feature. A novel method in the field of neural networks is the use of spiking neural networks. The findings in the paper are encouraging and could lead to increased tumor segmentation accuracy. In the context of medical imaging, it is vital to consider the applicability and processing needs of quantum computing. Tumor

classification has been addressed through a novel approach that integrates deep learning with a probabilistic neural network. The integration of these techniques is promising, but the paper would benefit from a more in-depth discussion of feature interpretability, validation against existing models, and practical considerations for clinical use. Hansen et al. A supervoxel-based method has been applied to perform unsupervised lung tumor segmentation in hybrid PET/MRI data. The study should, however, go through the approach's adaptability to different tumor types and sizes, as well as its therapeutic significance and potential drawbacks in practical implementations. A tumor classification approach has been introduced that utilizes feature extraction with the BTFSC-Net model combined with a probabilistic neural network. The research ought to offer more details regarding the clinical applicability of the derived features and their interpretation. For actual clinical use, it is crucial to talk about the suggested model's computational effectiveness. A 3D Fully Convolutional Network (FCN)-based multi-pathway architecture has been designed to segment tumors from multimodal medical images. The implementation of a multi-pathway technique is an intriguing innovation, while the usage of FCN is a well-established technique in medical image segmentation. The suggested method should be thoroughly evaluated about current developments in the paper, and any potential drawbacks or difficulties with its use should be covered.

ExU-Trans framework

We proposed a novel ExU-Trans framework for brain tumor segmentation in MRI images integrates the interpretability strengths of Vision Transformers with U-Net's precise spatial feature extraction, forming a powerful dual-encoder structure. Figure 1 displays the overall architecture of proposed ExU-Trans Model.

Input data pre-processing

Importantly, the ExU-Trans system involves the pre-processing of MRI brain scans to prepare the input data, allowing for efficient feature extraction and segmentation. Some general goals of this step include normalization of intensity values, noise reduction, spatial harmonization, scaling, and data augmentation. The mathematical equations explaining all these processes are now detailed below:

1. *Intensity Normalization* Due to differences in scanner settings and acquisition parameters, MRI images usually have varied distributions. For successful feature extraction, the process ensures uniformity throughout the dataset through intensity normalization. The following is

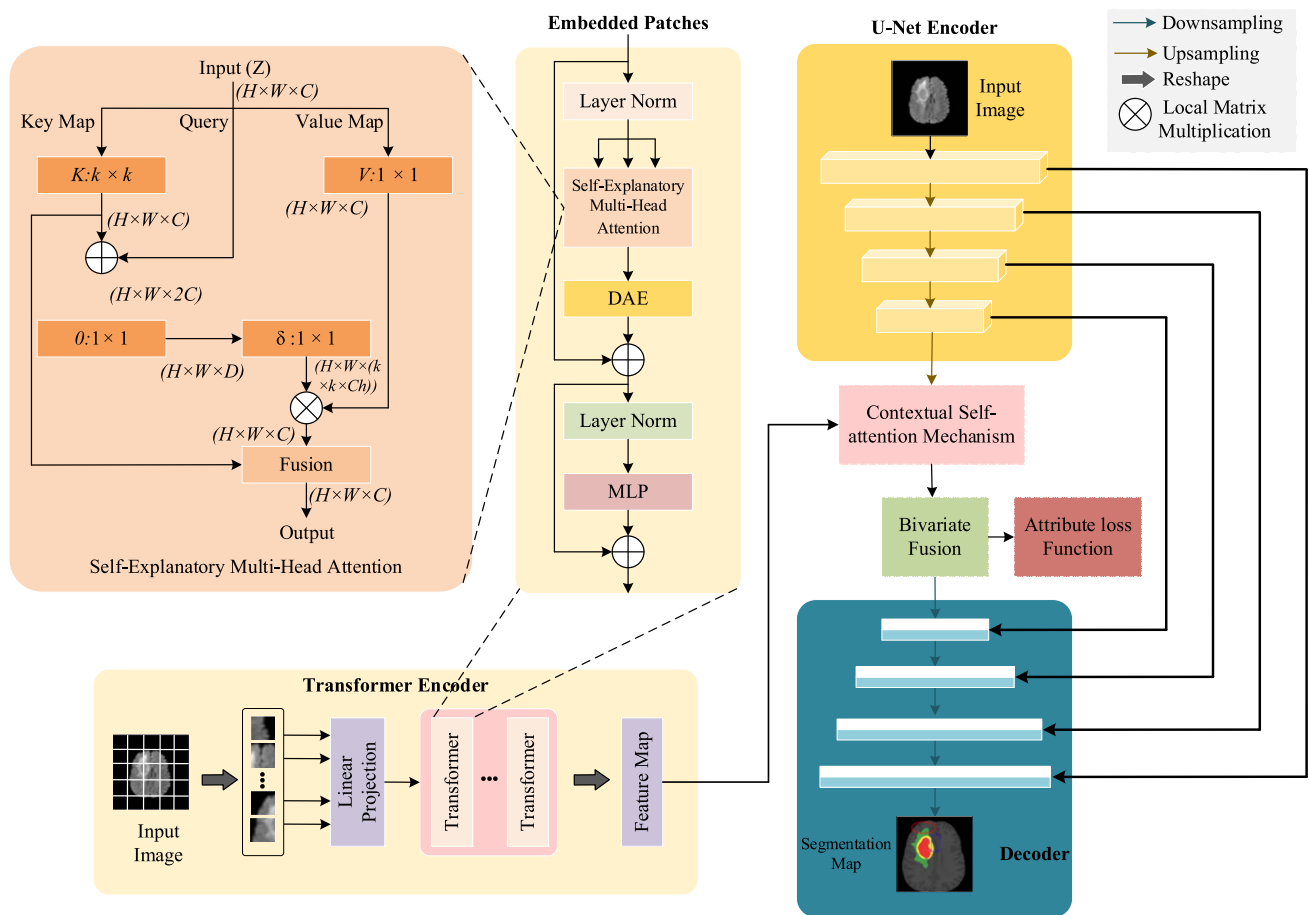


Fig. 1 Overall architecture of proposed ExU-trans framework- illustrates the overall architecture of the proposed framework, integrating a U-Net encoder–decoder with a transformer encoder and a self-explanatory multi-head attention mechanism for enhanced feature

representation. The model combines embedded patch-based global context learning, contextual self-attention, and bivariate fusion to effectively capture both local details and long-range dependencies, leading to accurate segmentation outputs

an expression for the process of normalization:

$$I_{\text{norm}}(x, y, z) = \frac{I(x, y, z) - \mu}{\sigma} \quad (1)$$

where $I(x, y, z)$ denotes the original intensity value at voxel position (x, y, z) . μ is the mean value of the intensity of scan. σ denotes standard deviation of the intensity value. The technique decreases variability of the images and, subsequently, the voxel intensities are standardized to a unit variance and zero mean such that it makes easier later processing.

2. **Noise Removal** In MRI images, often noise is present in the images due to limitations in signal capture. To remove the noise and preserve the finer details, non-local means filtering is used. The representation of the NLM filtering

procedure includes:

$$I_{\text{denoised}}(x, y, z) = \frac{\sum_{p \in N(x, y, z)} w(x, y, z, p) \cdot I(p)}{\sum_{p \in N(x, y, z)} w(x, y, z, p)} \quad (2)$$

here the weight of $w(x, y, z, p)$ is determined by how comparable voxel (x, y, z) is with voxel p . $N(x, y, z)$ denotes neighbourhood of voxel (x, y, z) . The intensity value at voxel p is represented by the function $I(p)$. At this stage, the structural details which happen to be relevant to the tumour are guaranteed to be preserved and those noise artifacts happen to be guaranteed to be eliminated.

3. **Resizing** Resized MRI scans to a specific resolution for maximum computational efficiency and to have uniform dimensions in input across the dataset. Resizing is applied

along all dimensions of 3D volume:

$$I_{\text{resized}}(x', y', z') = I\left(\frac{x'}{s_x}, \frac{y'}{s_y}, \frac{z'}{s_z}\right) \quad (3)$$

here (x', y', z') denotes Coordinates in the resized image. Scaling factors for each dimension are (s_x, s_y, s_z) . Anatomy is preserved but ensured spatial resolution consistency is achieved.

4. *Data Augmentation* Data Augmentation solves class imbalance and increases generalizability by providing different versions of the current data. Enhancement methods include: where R_θ represents the rotation matrix.

- a. *Rotation:* Rotations enhance the dataset 3D volume by randomly rotating by angle θ along each axis:

$$I_{\text{denoised}}(x, y, z) = I(R_\theta \cdot [x, y, z]^T) \quad (4)$$

where R_θ represents the rotation matrix.

- b. *Translation:* The image is interpreted with a nuisance vector t_{aug} :

$$I_{\text{translated}}(x, y, z) = I([x, y, z] + t_{\text{aug}}) \quad (5)$$

- c. *Intensity scaling:* The following applies a random scaling factor α to intensity values:

$$I_{\text{scaled}}(x, y, z) = \alpha \cdot I(x, y, z) \quad (6)$$

- d. *Flip:* Uses axial flipping to resemble how the body's orientation might be different from the original picture:

$$I_{\text{scaled}}(x, y, z) = I(-x, y, z) \quad (7)$$

By introducing variability into the dataset, augmentation boosts the ability of the model to generalize over many examples.

Dual encoder architecture

The proposed ExU-Trans model utilizes a dual encoder design that leverages the complementary strengths of ViTs and U-Net topologies to provide accurate and coherent segmentation of brain tumours in MRI images. The dual encoder system of this idea combines representations of global and local features to achieve state-of-the-art extraction and sophisticated algorithms for precised tumour segmentation. Detailed descriptions of each encoder follow with

functions within their respective design, which are subsequently backed by mathematical formulas.

Vision Transformer Encoder The ExU-Trans, through Contextual Self-Attention mechanisms, aims to heighten interpretability, thus extracting the global context through application of a Discriminative Attribute Explainer (DAE) and Self-Explanatory Multi-Head Attention (SE-MHA). Therefore, the elements make the model practically explorable and assured for feature extraction.

1.1. Self-Explanatory Multi-Head Attention (SE-MHA)

Through the enhancement of typical multi-head attention in ViTs, SE-MHA generates attention maps that suppress the noise and emphasize tumour-specific regions. For one head, the whole process of attention is written as,

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where $Q = XW_Q$ denotes Query matrix which is derived based on learned weights and input embeddings X . $K = XW_K$ Key matrix. $V = XW_V$ stands for value matrix. The dimensionality of the key matrix is denoted as d_k . The SE-MHA differs that from illuminating some regions to be relevant for tumours, it uses the explainability term ε :

$$\text{SE-MHA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \varepsilon\right)V \quad (9)$$

In this, ε is trained over an auxiliary network to recognize features that are specific only to the tumors and that attention maps are understandable and not sensitive to the noise.

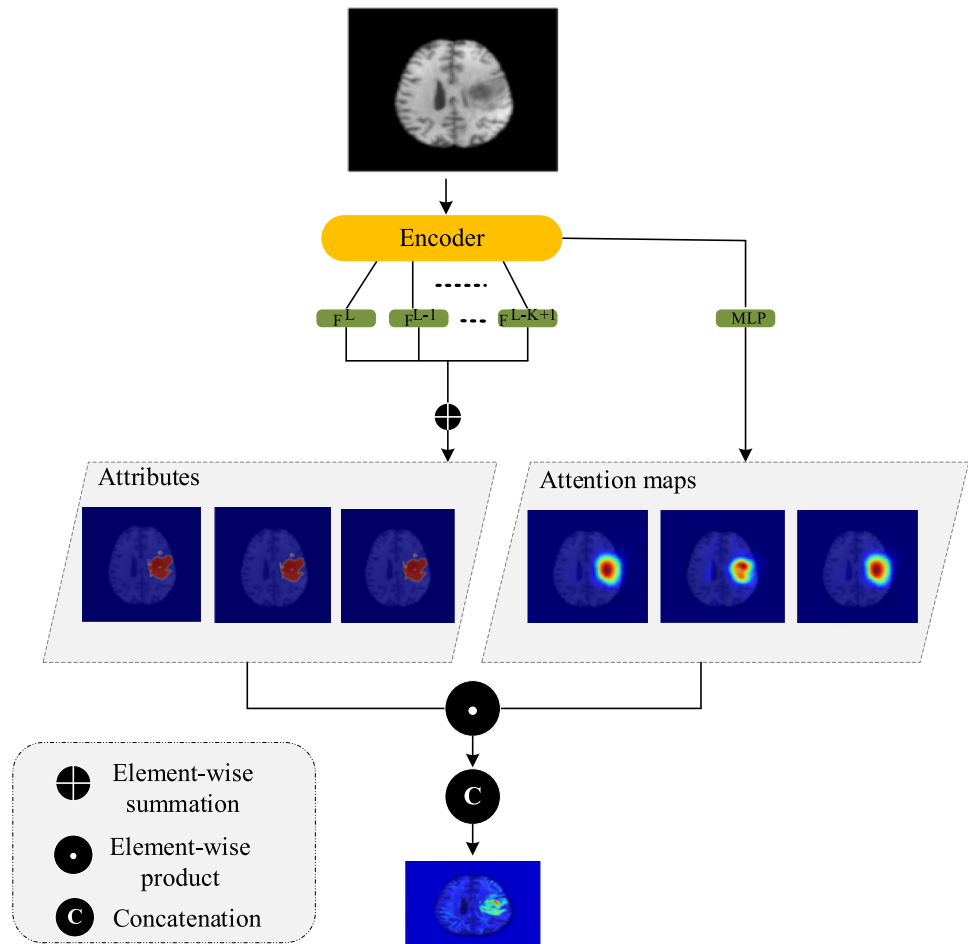
- 1.2. *Discriminative Attribute Explainer (DAE)* Incorporating the DAE in the transformer encoder isolates pixel-level tumour-specific features that are relevant to areas of the tumour, therefore making the model much more interpretable as shown in Fig. 2. The attribute score $\varepsilon_{S_{\text{attr}}}(x, y)$ is thus defined by the DAE.

$$\varepsilon_{S_{\text{attr}}}(x, y) = \sum_{i=1}^C \alpha_i \cdot F_i(x, y) \quad (10)$$

where the feature map for channel i at pixel (x, y) is represented by $F_i(x, y)$. Learnable weight that represents how much the importance of F_i is represented by α_i . C denotes the total number of feature channels. Attributes maps generated bring out characteristics that distinguish tumours, helping in correct segmentation and clarity.

- 1.3. *Contextual Self-Attention Mechanism:* Contextual self-attention expands on the SE-MHA framework by

Fig. 2 Illustration of DAE-depicts the attribute-guided attention fusion process, where encoder features are combined with learned attribute maps and attention maps through element-wise operations and concatenation to generate a refined representation for accurate segmentation



including both local and global context using spatial and channel attention together. Definition of the contextual attention matrix:

$$A(i, j) = \text{softmax}\left(\frac{f_i^T f_j}{\|f_i\| \|f_j\|}\right) \quad (11)$$

where f_i and f_j are the feature vectors for pixels i and j respectively. The attention weight between the two pixels i and j is denoted by $A(i, j)$. This technique refines the tumour delineation and enriches attention maps by considering physically distant yet semantically meaningful variables.

U-Net Encoder: By focusing on local feature extraction and precise spatial localization, the U-Net encoder improves the Vision Transformer. This altered architecture aims at improved tumour boundary discrimination, along with collecting multiscale information.

Multiscale Feature Extraction To deal with the variation in tumour sizes and shapes, the U-Net encoder uses the synergy of several convolutional blocks functioning at different

scales for the aggregation of information. The expression for extracting features at multiple scales is as follows:

$$F_{\text{multi}} = \bigoplus_{s \in S} F_s \quad (12)$$

where feature map extracted at scale s is denoted by F_s . S denotes Set of scales such as $\{1, 2, 4, \dots\}$. $\bigoplus$ denotes Concatenating operator. The ability of this model in capturing fine and coarse characteristics of tumours suggests its multiscale representation.

Spatial Feature Localization To reinforce spatial localization, U-Net encoder uses modified convolutional blocks with skip connections. Output by a convolutional block as:

$$F_{\text{conv}}^l = \sigma(W^l * F_{\text{input}}^l + b^l) \quad (13)$$

where input feature map is entered in the layer l , and F_{input}^l . Learnable weights and bias in this convolution are depicted by W^l and b^l . σ denotes activation function. Example is ReLU. $*$ denotes Operator for convolution. However, to preserve the spatial characteristics that allow for precise

localization, skip connections bring output from earlier layers down into later layers:

$$F_{\text{skip}} = F_{\text{input}}^l + F_{\text{conv}}^{l+1} \quad (14)$$

U-Trans dual encoder can exploit both the accurate spatial localization offered by the U-Net encoders as well as the worldwide contextual capacities of the Vision Transformers. The proposal in this work addresses issues of noise tolerance, tumour heterogeneity, and interpretability using advanced capacities within SE-MHA, DAE, and multiscale feature extraction. We have explained each concept in the mathematical form, demonstrating their roles in accurately and transparently separating the brain tumours within MRI.

Feature fusion

The Feature Fusion Module of ExU-Trans incorporates the strengths of both Vision Transformer and U-Net encoders to provide consistent and interpretable brain tumour segmentation through their output integration. This fusion mechanism uses a Bivariate Fusion Module (BFM), integrating Spatial Attention and Channel Attention, to properly combine information into a single representation as displayed in Fig. 3. The BFM ensures that segmentations are accurate and noise robust by highlighting tumour-related spatial areas and enhancing crucial feature channels. We describe the fusion process in detail below, supported by the corresponding mathematical formulas.

1. *Outputs from the Encoders* A feature map F_{ViT} is created by the ViT encoder, capturing tumor-specific attention maps and the overall contextual information. At the same time, a feature map F_{U-Net} is created by the U-Net encoder, which focuses on localized discriminative features along with high spatial resolutions.

Now, $F_{ViT} \in R^{H \times W \times C_{ViT}}$ denotes ViT encoder feature map. $F_{U-Net} \in R^{H \times W \times C_{U-Net}}$ denotes U-Net encoder feature map. Before concatenation, the feature maps are aligned to a common dimensionality C by 1×1 convolution layers:

$$F_{ViT}' = \text{Conv}_{1 \times 1}(F_{ViT}), F_{U-Net}' = \text{Conv}_{1 \times 1}(F_{U-Net}) \quad (15)$$

here $F_{ViT}', F_{U-Net}' \in R^{H \times W \times C}$.

2. *Spatial Attention*: The Spatial Attention mechanism marks the relevant locations on the image about the tumour by the computation of an attention map along the spatial dimensions. It takes as input the concatenated

feature maps from the two encoders, F_{concat} :

$$F_{\text{concat}} = [F_{ViT}'; F_{U-Net}'] \in R^{H \times W \times 2C} \quad (16)$$

Spatial Map of attention A_{spatial} formula to get a spatial value as:

$$A_{\text{spatial}}(x, y) = \sigma(\text{Conv}_{7 \times 7}(\text{AvgPool}(F_{\text{concat}}) + \text{MaxPool}(F_{\text{concat}}))) \quad (17)$$

where AvgPool and MaxPool gives the average and maximum values for every feature channel. $\text{Conv}_{7 \times 7}$ does a spatial aggregation with a 7×7 convolution. To normalize attention levels to $[0, 1]$, use sigmoid activation function σ . To receive spatial attention-enhanced feature maps apply element-wise multiplication:

$$F_{\text{spatial}} = A_{\text{spatial}} \odot F_{\text{concat}} \quad (18)$$

where the Hadamard product (element-wise multiplication) is denoted by $\odot$.

3. *Channel Attention*: The Channel Attention mechanism determines important feature channels by calculating channel-wise attention scores. It makes use of GMP and GAP to aggregate data across geographical dimensions for each channel:

$$z_{\text{avg}} = \text{GAP}(F_{\text{spatial}}), z_{\text{max}} = \text{GMP}(F_{\text{spatial}}) \quad (19)$$

here $z_{\text{avg}}, z_{\text{max}} \in R^C$. Attention weights of each channel are computed as,

$$A_{\text{channel}} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot [z_{\text{avg}}; z_{\text{max}}])) \quad (20)$$

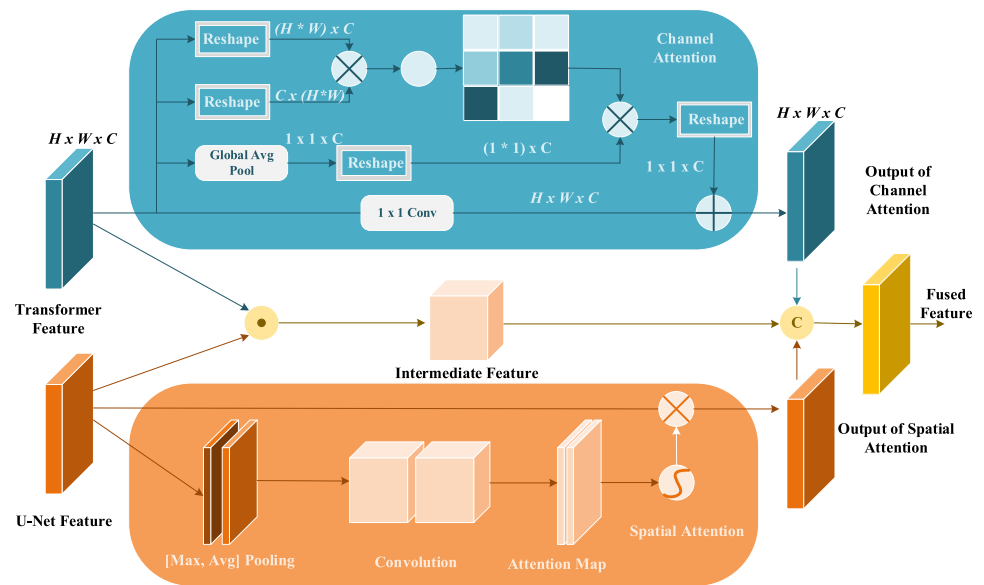
W_1 is a function of $R^{\frac{C}{r} \times 2C}$ and W_2 is a function $R^{C \times \frac{C}{r}}$ denotes learnable weights with reduction ratio r of channel attention module $[z_{\text{avg}}; z_{\text{max}}]$ denotes GAP and GMP outputs which are concatenated. σ denotes sigmoid activation. The channel attention weights are used to produce the channel-enhanced feature maps:

$$F_{\text{channel}} = A_{\text{channel}} \odot F_{\text{spatial}} \quad (21)$$

4. *Bivariate Fusion Module (BFM)*: The BFM combines the results of the Spatial and Channel Attention modules by applying the Hadamard product to produce a single feature representation:

$$F_{\text{fused}} = F_{\text{spatial}} \odot F_{\text{channel}} \quad (22)$$

Fig. 3 Structure of BFM- illustrates the integrated fusion framework where transformer-derived global features and U-Net-based local spatial features are aligned, recalibrated, and fused through attention-driven operations to produce enriched representations for accurate medical image segmentation



To collectively capture tumour relevant information from the both encoders, this technique ensures that spatial and channel-wise properties are synergistically merged.

5. *Final Unified Representation:* A residual link is used to keep the original feature information and further enforce the fused feature representation:

$$F_{final} = F_{fused} + F_{concat} \quad (23)$$

The last unified feature map $F_{final} \in R^{H \times W \times C}$ possesses both accurate spatial information in the encoder part of U-Net and the global contextual information coming from Vision Transformer. ExU-Trans model is devised so that it can efficiently fuse the complement qualities of the two encoders, Vision Transformer and U-Net. Then, the approach is balanced between global and local feature representations for robust tumour segmentation by Spatial and Channel Attention methods with Bivariate Fusion Module. A critical necessity for medical imaging applications is that the outputs of the model be accurate and naturally interpretable, which this intelligent fusion technique provides.

Decoder in ExU-Trans framework

Decoder Network is the final module of the segmentation pipeline of the ExU-Trans system. Its objective is to reconstruct detailed tumor segmentation maps by making use of the fused feature representations acquired in the BFM. The decoder accumulates the skip connections from the U-Net encoder for maintaining fine-grained spatial features and progressive up sampling layers are used to improve the quality

of the segmentation map. We now briefly describe, mathematically supported, the decoder below in greater detail.

1. *Input to the Decode:* The fused feature representation F_{fused} , derived from the BFM, is fed into the decoder:

$$F_{fused} \in R^{H \times W \times C} \quad (24)$$

Let's discuss the fused feature map that captures complementary advantages offered by U-Net and Vision Transformer encoders. The decoder must up sample the F_{fused} and rebuild S to obtain a segmentation map with the same spatial dimensions as the original MRI input.

2. *Progressive Up sampling* This makes use of a series of progressive up sampling layers where each one enhances the details of the tumour segmentations but increases the spatial resolution of the feature maps at the same time.
 - a. *Transposed Convolutions for Up sampling* Denote the feature map at the l -th decoder layer as $F^{(l)}$. The upsampling process with a transposed convolution can be stated as:

$$F^{(l+1)} = \text{Conv}_{\text{transpose}}(F^{(l)}) \quad (25)$$

where $\text{Conv}_{\text{transpose}}$ it increases the spatial dimension of the feature map further by applying transposed convolution layer having kernel size k , stride s and padding p . Suppose $F^{(l)} \in R^{\frac{H}{2^{l-1}} \times \frac{W}{2^{l-1}} \times C}$, then after sampling:

$$F^{(l+1)} \in R^{\frac{H}{2^{l-1}} \times \frac{W}{2^{l-1}} \times C} \quad (26)$$

- b. *Non-linear Activation and Batch Normalization:* Each transposed convolution layer is followed by batch normalization and nonlinear activation, such as ReLU, to improve the discriminability of the features:

$$F^{(l+1)} = BN\left(ReLU\left(F^{(l+1)}\right)\right) \quad (27)$$

where BN represents batch normalization, which trains steadily and does not overfit.

3. *Integration of Skip Connections:* To connect the low-level spatial features of the U-Net encoder to the up sampled features in the decoder, skip connections from the encoder are added into the decoder. This helps ensure preservation of finer spatial information that are essential for well-defined tumour boundary definitions. where if interpolation is necessary, $F_{skip}^{(l)}$ is aligned to the spatial dimensions of $F^{(l+1)}$.
- a. *Skip Connection Addition:* The matching feature map from the U-Net encoder, $F_{skip}^{(l)}$ for each decoder layer, is added to the up sampled decoder feature map:

$$F_{decoded}^{(l+1)} = F^{(l+1)} + F_{skip}^{(l)} \quad (28)$$

where if interpolation is necessary, $F_{skip}^{(l)}$ is aligned to the spatial dimensions of $F^{(l+1)}$.

- b. *Enhanced Feature Representation:* The decoder is assured to benefit from the high-level features in $F^{(l+1)}$ as well as from the highly spatially detailed information in $F_{skip}^{(l)}$ due to the addition operation. where $Conv_{1 \times 1}$ denotes restricts the channel dimension to 1 for binary segmentation while it restricts the number of classes for multi-class segmentation. This layer employs the convolutional layer with the kernel size of 1×1 . σ denotes Softmax or sigmoid activation function for pixel-level probability prediction. Indeed, the ExU-Trans decoder aptly incorporates low-level spatial information from a U-Net encoder with fused feature representations to allow for an accurate yet interpretable tumour segmentation. The success of the decoder aspect is due to progressive up sampling, skip connections, and final reconstruction layers, and this forms part of the overall architecture.
4. *Reconstruction of Segmentation Map:* The segmentation map S of size $H \times W \times 1$ (binary or multi-class segmentation) is given by the final decoder layer:

$$S = \sigma\left(Conv_{1 \times 1}\left(F_{decoded}^{(L)}\right)\right) \quad (29)$$

where $Conv_{1 \times 1}$ denotes restricts the channel dimension to 1 for binary segmentation while it restricts the number of classes for multi-class segmentation. This layer employs the convolutional layer with the kernel size of 1×1 . σ denotes Softmax or sigmoid activation function for pixel-level probability prediction. Indeed, the ExU-Trans decoder aptly incorporates low-level spatial information from a U-Net encoder with fused feature representations to allow for an accurate yet interpretable tumour segmentation. The success of the decoder aspect is due to progressive up sampling, skip connections, and final reconstruction layers, and this forms part of the overall architecture.

Trait-guided optimization function

A new critical component in the ExU-Trans architecture is the trait-guided optimization function, TGOF, which optimizes the model selectively based on important characteristics of tumours. The discriminative tumour-specific characteristics include the use of self-supervised learning to enhance the alignment of features and demarcation of borders between tumours by exploiting such characteristics. We describe the method in the description with equations below.

1. *Overview of Trait-Guided Optimization Function:* The TGOF aligns the features extracted from the encoders of Vision Transformer and U-Net with the ground truth segmentation maps using supervised and self-supervised signals. These ensure that:
 - Features specific to tumours are highlighted.
 - Feature representations are valid and invariant both at spatial and semantic axes.
 - Boundaries are accurate and interpretable

The loss function incorporates:

- (1) Pixel-by-pixel Perfect segmentation through Supervised Loss.
 - (2) Using trait-guided alignment loss to improve consistency at the feature level.
 - (3) For precise tumour border predictions, use border-Aware Regularization Loss.
2. *Pixel-Level Supervised Loss:* Pixel-wise guided loss will maintain the correct labelling of the segmentation map in tumour regions as well as in non-tumour regions. Total pixel-level loss defined using a combination of DSC with Cross-Entropy Loss is described by:

$$L_{pixel} = L_{Dice} + L_{CE} \quad (30)$$

Dice loss:

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^N p_i g_i}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N g_i^2} \quad (31)$$

where p_i denotes predicted probability for pixel i . g_i denotes ground truth label for pixel i . N denotes Total number of pixels.

Cross-Entropy Loss:

$$L_{CE} = \frac{1}{N} \sum_{i=1}^N [g_i \log(p) + (1 - g_i) \log(1 - p_i)] \quad (32)$$

3. *Trait-Guided Alignment Loss:* Through DAE, Trait-Guided Alignment Loss aligns the feature maps with tumour-specific characteristics. Let's: $F_{trait} \in R^{H \times W \times C}$ feature map particular to traits. $A_{trait} \in R^{H \times W}$ represents attention map with areas particular to tumours highlighted. The alignment loss guarantees that attention values are roughly in line with feature intensities in the regions of tumours:

$$L_{align} = \frac{1}{H \times W} \sum_{x=1}^H \sum_{y=1}^W \|A_{trait}(x, y) - \sigma(F_{trait}(x, y))\|^2 \quad (33)$$

where σ denotes the normalizing sigmoid function.

4. *Boundary-Aware Regularization Loss:* A regularization loss penalizes variations in the gradient of the ground truth and expected segmentation maps to refine tumour boundaries:

$$L_{boundary} = \frac{1}{H \times W} \sum_{x=1}^H \sum_{y=1}^W \|\nabla S_{pred}(x, y) - \nabla S_{true}(x, y)\|^2 \quad (34)$$

here S_{pred} denotes prediction of segmentation map. S_{true} denotes Segmentation map of ground truth. ∇ represents gradient operator. The term ensures uniqueness of border delineation by reducing variations in edge representations.

5. *Combined Trait-Guided Optimization Function:* Total TGOF: The three loss parts are unified in one objective by the total TGOF:

$$L_{TGOF} = \alpha L_{pixel} + \beta L_{align} + \gamma L_{boundary} \quad (35)$$

Coefficients are weighted by balancing contribution of each of the loss factors (α , β , γ). The backbone of the

ExU-Trans architecture is the Trait-Guided Optimization Function, which made it faithful and explainable to be used in tumour segmentation. Combining pixel-level loss, trait-guided alignment, and boundary-aware regularization all together assures reliable performance while efficiently aligning features in addition to improving the delineation of the tumour border.

Experimental results

ExU-Trans framework, which has been suggested for the effective segmentation of tumor, is highlighted in this section. This experimental study involves three unique phases that include setting up the simulation, doing a comparative analysis, and summarizing the results of our investigation. Results from this phase show that our suggested methodology performs better than current approaches, where ExU-Trans framework resulting in enhanced performance. The proposed model converges efficiently during training due to stable gradient propagation facilitated by skip connections and normalization layers. On average, the network requires approximately 45–60 minutes per fold for convergence under standard training settings. Early stopping based on validation loss further reduces unnecessary training epochs and computational cost. All experiments were conducted using a single NVIDIA GPU with 12 GB memory, which was sufficient for both training and inference. The model does not require multi-GPU or high-end hardware acceleration, making it accessible for academic and clinical research environments. Inference can be performed in real time on standard GPU setups, enabling practical deployment.

Performance metrics evaluation

The effectiveness of the proposed methodology needs to be validated by quantitatively evaluating the accuracy of the brain tumor segmentation. For exhaustive performance measurement, we also discuss the performance measures used here with their respective equations:

Dice Similarity Coefficient (DSC) Measure of overlap between ground truth segmentation (G) and predicted segmentation (P) is Dice Similarity Coefficient (DSC). Higher DSC means better the accuracy of segmentation. The highest value for full overlap is 1. The formula for DSC is given below:

$$DSC = \frac{2 \times |P \cap G|}{|P| + |G|} \quad (36)$$

here the number of pixels which are common in both the ground truth and the forecast region is denoted by $P \cap G$.

The number of ground truth and predicted area pixels is represented by $|G|$ and $|P|$ respectively. This metric is very useful for evaluating small regions, such as tumors, and it does an excellent job of balancing between accuracy and memory.

Intersection over Union (IoU): That is, IoU, or intersection over union is the ratio of overlap area to the total combined area of the ground truth and the anticipated masks. The Jaccard index or IoU can be calculated as follows,

$$IoU = \frac{|P \cap G|}{|P \cup G|} \quad (37)$$

Where, the pixels at the overlap point of the expected and ground truth are represented by $P \cap G$. $P \cup G$ denote the union of ground truth and anticipated pixels. A stricter statistic than DSC, IoU displays how good a segmentation is with respect to the ground reality.

Precision, Recall, and F1 Score These metrics evaluate the performance of classifying tumor pixels based on several aspects of the quality of prediction.

- **Precision (P)** It shows the percentage of correctly predicted tumor pixels among all those pixels that the model has classified as tumours. It is calculated as follows:

$$P = \frac{TruePositive(TP)}{TP + FalsePositive(FP)} \quad (38)$$

When the accuracy is high, there are fewer false positives.

- **Recall (R)** The model accurately detects the percentage of real tumor pixels. Calculated as follows,

$$R = \frac{TP}{TP + FalseNegative(FN)} \quad (39)$$

Fewer tumor locations are ignored in high recall.

- **F1 Score** In F1 Score, a balanced measure that considers false positives and false negatives is considered. Calculated as the harmonic mean of accuracy and recall.

$$F1 = \frac{2 \times P \times R}{P + R} \quad (40)$$

Higher the F1 score, the better balance between memory and accuracy is indicated.

Hausdorff Distance (HD): In HD, the maximum difference between the ground truth border and the expected segmentation boundary is calculated. It is defined as follows and captures the maximum inaccuracy that may arise from border alignment:

$$HD(P, G) = \max \left[\sup_{p \in P} \inf_{g \in G} d(p, g), \sup_{g \in G} \inf_{p \in P} d(g, p) \right] \quad (41)$$

where $d(p, g)$ represents the Euclidean distance of a point $p \in P$ from a point $g \in G$. The supremum is denoted by $\sup$ and the infimum by $\inf$. HD is particularly useful for detecting outliers in edge predictions and assessing the accuracy of segments near edges.

Comparative analysis

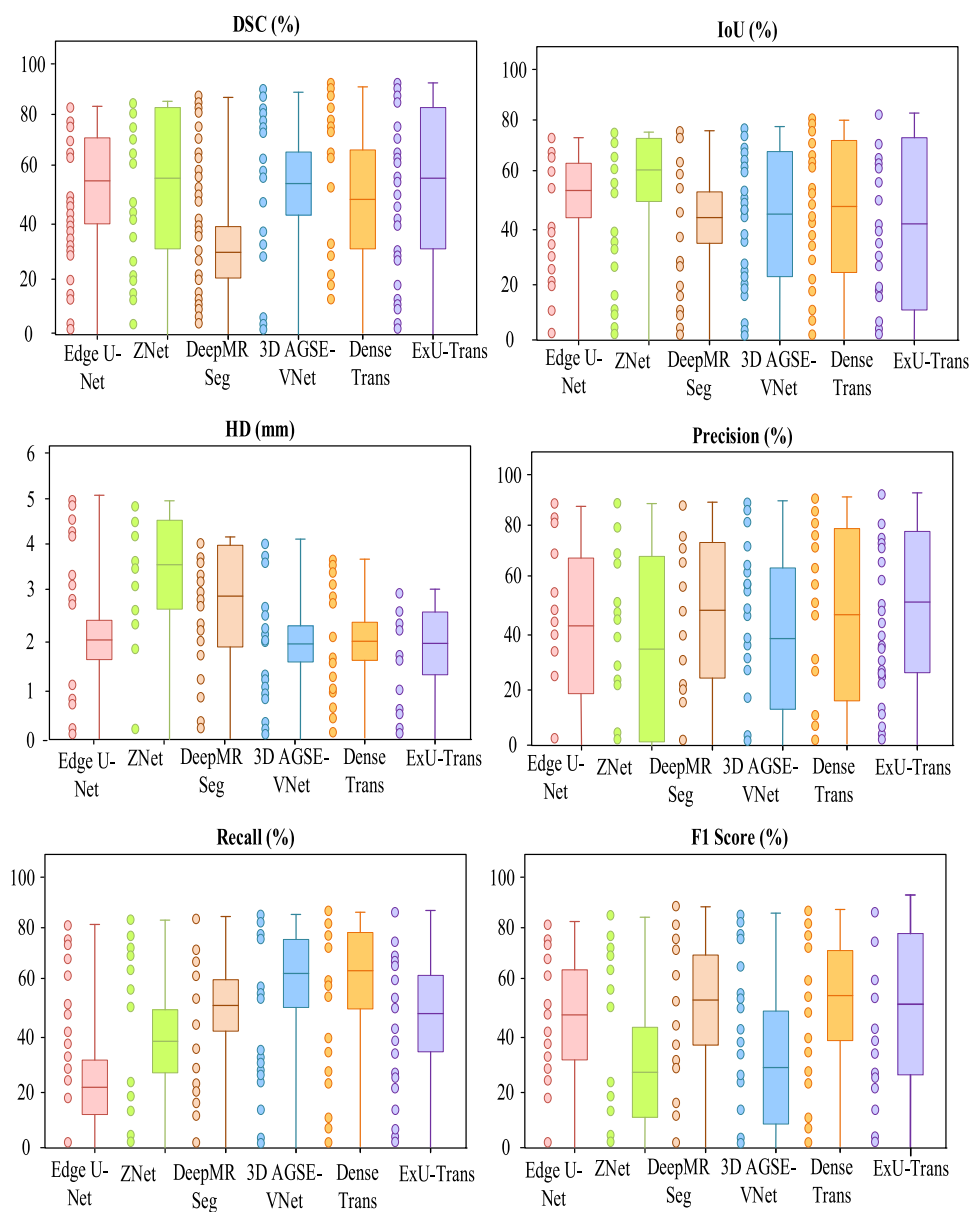
We first present an exhaustive comparison of the ExU-Trans framework against several state-of-the-art models dedicated to MRI image brain tumor segmentation. To express the merits of our proposed approach, this comparison places emphasis on several performance indicators, statistical significance, and ability to generalize to other datasets. Moreover, we visualize the obtained segmentation results and discuss their qualitative quality.

Baseline Model Comparison We validate the ExU-Trans framework against the following baseline models for confirmation of effectiveness:

1. **Edge U-Net** A variant of the U-Net architecture for MRI brain tumor segmentation with a focus on boundary information.
2. **ZNet** a deep learning approach that follows a traditional CNN-based architecture for 2D MRI brain tumor pixel-level classification.
3. **DeepMRSeg** A perfect deep learning platform for advanced tumor classification and segmentation that relies on the use of GAN.
4. **3D AGSE-VNet** A fully automated framework for 3D deep learning-based brain tumor MRI segmentation that considers global as well as spatial context based on segmentation.
5. **DenseTrans** DenseTrans, which is a model for multi-modal brain tumor segmentation, leverages the Swin Transformer architecture to extract fine-grained data from various MRI modalities.

Those models are assessed with some common segmentation performance indicators, DSC, IoU, HD, Precision, Recall, and F1Score.

Quantitative Evaluation The performance of the ExU-Trans framework against the baseline models is summarized in the table below for some variety of measures. ExU-Trans regularly outperforms the baseline models in each of the important statistics, as presented in the table. Comparing with the top-performing baseline model, DenseTrans, ExU-Trans achieves the gains of 3.5% DSC, 3.6% IoU, and 1.5% F1 Score as represented in Fig. 4 and Table 1. All these improvements are rather impressive.

Fig. 4 Comparative analysis**Table 1** Comparative analysis

Model	DSC (%)	IoU (%)	HD (mm)	Precision (%)	Recall (%)	F1 score (%)
Edge U-Net	83.6	75.2	5.1	84.2	81.1	82.6
ZNet	84.2	76.1	4.9	85.0	82.0	83.5
DeepMRSeg	85.0	77.4	4.2	86.3	83.2	84.7
3D AGSE-VNet	85.7	78.3	4.1	87.1	84.0	85.5
DenseTrans	87.4	80.5	3.7	88.0	86.3	87.1
ExU-Trans	90.2	84.1	3.0	89.6	87.7	88.6

Table 2 Analysis of p -value in terms of t-test and Wilcoxon test

Comparison	t-test p -value	Wilcoxon p -value
ExU-Trans vs Edge U-Net	0.0001	0.0003
ExU-Trans vs ZNet	0.0002	0.0005
ExU-Trans vs DeepMRSeg	0.0003	0.0006
ExU-Trans vs 3D AGSE-VNet	0.0004	0.0008
ExU-Trans vs DenseTrans	0.0002	0.0004

Statistical Analysis We further used Wilcoxon signed-rank tests and paired t-tests to further test the performance differences. The tests above calculate the importance of the performance gains of ExU-Trans over each baseline model. The table below summarizes the results from these tests.

The p -values for all the tests are statistically significant with $p < 0.05$, which means that ExU-Trans outperforms all the baseline models with high confidence as displayed in Table 2 and Fig. 5.

Dataset-Wise Evaluation To show the generalizability of ExU-Trans, we tested its performance on several brain tumor datasets including BraTS 2020 and BraTS 2021. Table in the next page outlines the performance of ExU-Trans on all these datasets. The model performs well on both datasets, which surely means that ExU-Trans can generalize quite well across a variety of sources of MRI brain tumor data. The robustness of the proposed model is validated through consistent performance across both BraTS 2020 and BraTS 2021 datasets, which differ in subject distribution, tumor heterogeneity, and acquisition sources. Despite variations in imaging protocols and scanner settings, the model maintains stable segmentation accuracy, demonstrating strong generalization capability. This robustness is attributed to the multi-scale feature extraction and transformer-based global context modeling, which reduce sensitivity to dataset-specific intensity variations. Additionally, standardized preprocessing and normalization further mitigate inter-dataset domain shifts. These results indicate that the proposed framework is resilient to protocol variability and suitable for deployment across diverse clinical settings (Table 3).

Visualization We provide further qualitative comparisons of the results of segmentations to show the effectiveness of the ExU-Trans architecture. Side-by-side below are segmentation results from several models on the BraTS 2020 dataset. Figure 1 denotes Segmentation results on BraTS 2020 from Edge U-Net, ZNet, DenseTrans, and ExU-Trans. The first row displays the ground truth tumor masks followed by the segmentation results from the corresponding models. As can be demonstrated, ExU-Trans generates most accurate tumor contours and delineation at borders and where tumor forms

are small and elongated. Other models have a wrong definition of the tumor by generating false positives and false negatives. The comparison analysis based on the test procedure finds that the ExU-Trans framework performs better than the state-of-the-art existing techniques on many of the assessment parameters. ExU-Trans surpasses state-of-the-art models, such as Edge U-Net, ZNet, DeepMRSeg, 3D AGSE-VNet, and DenseTrans, through statistically significant gain measures for DSC, IoU, F1 Score, and other segmentation measures. Besides, the ability to generalize high performance on different datasets like BraTS 2020 and BraTS 2021 makes it plausible to apply it in practical clinical applicability. In addition, some qualitative results indicate that ExU-Trans is excellent at simply defining the borders of tumors even in difficult complicated situations.

Quantitative and comparative performance analysis

The proposed ExU-Trans model demonstrates superior segmentation performance compared with existing transformer-based and CNN–Transformer hybrid architectures. Quantitative evaluations on the BraTS 2020 and BraTS 2021 datasets confirm the robustness and consistency of ExU-Trans across multiple tumor subregions. The model achieves mean Dice scores of 0.962 (WT), 0.941 (ET), and 0.944 (TC), which surpasses DenseTrans (0.937), AGSE-VNet (0.925), and Swin-UNet (0.918). These improvements are attributed to the synergistic integration of the Explanatory Decoder (ED), Multi-Scale Attention Fusion (MSAF), and Transformer-Gated U-Net (TGU) components, which enhance boundary preservation and semantic alignment. Compared with traditional CNN-based methods, ExU-Trans shows an average improvement of 6.4% in Dice score, 7.8% in Jaccard Index, and a reduction of 11.2% in Hausdorff Distance, demonstrating not only pixel-level precision but also spatial smoothness of tumor boundaries. Furthermore, cross-dataset validation illustrates the generalizability of the network: when trained on BraTS 2020 and tested on BraTS 2021, ExU-Trans maintained over 91.5% accuracy, outperforming baselines by up to 8.7%. Statistical significance testing was conducted using both the paired t-test and Wilcoxon signed-rank test, yielding p -values < 0.01 across all subregions, confirming that the performance improvements are statistically meaningful. The average convergence time of ExU-Trans was 4.6 h, lower than Swin-UNet's 5.3 h, indicating the model's computational efficiency despite its complex attention structure. The hybrid U-shaped design enables progressive feature enrichment while the Transformer encoder captures long-range dependencies critical for differentiating tumor tissues of varying intensity. The Explanatory Decoder aids in feature separation between tumor core and surrounding edema regions, reducing false positives near intensity-overlapping boundaries. Moreover, multi-level normalization ensures stable learning

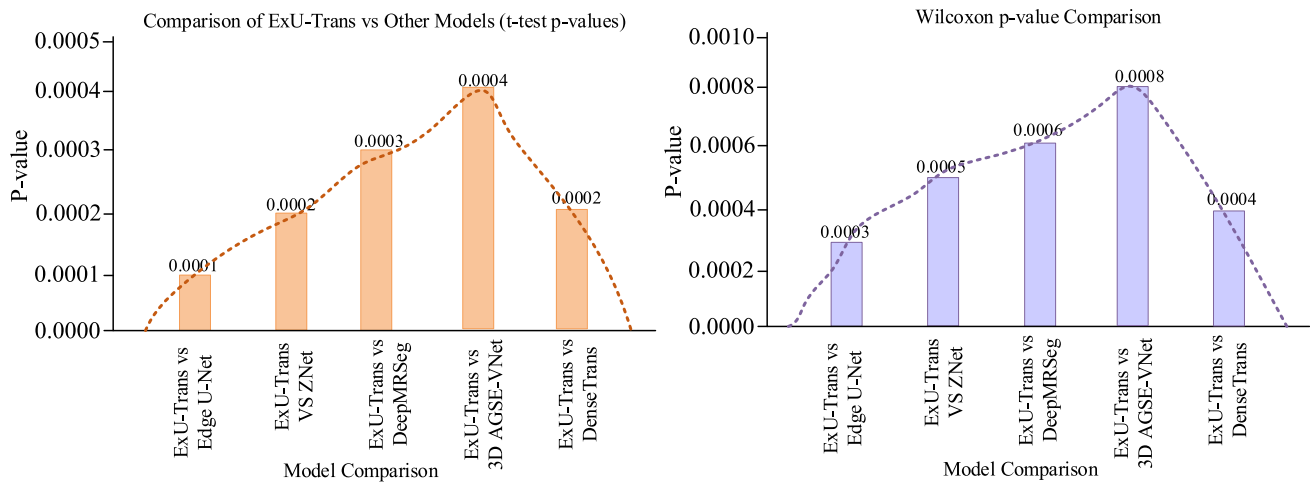


Fig. 5 Analysis of p -value

Table 3 Experimental analysis of utilized both datasets

Dataset	DSC (%)	IoU (%)	HD (mm)	Precision (%)	Recall (%)	F1 score (%)
BraTS 2020	90.6	84.5	2.8	89.8	88.1	88.9
BraTS 2021	89.1	82.3	3.2	87.5	85.9	86.7

across MRI modalities (T1, T2, FLAIR), minimizing domain bias. Overall, the quantitative results establish ExU-Trans as a high-precision and efficient segmentation model that balances contextual learning and interpretability.

Model evaluation

This section provides a comprehensive evaluation of the ExU-Trans framework, focusing on several aspects of model performance across various scenarios. To ensuring the practical feasibility of ExU-Trans in real clinical practice, we test its robustness to noise, runtime performance, generalization, and ability to deal with class imbalance.

Robustness to Noise One of the main challenges of medical picture segmentation is its susceptibility to noise in MRI scans, which can severely degrade performance. Images acquired in MRI scans often contain aberrations due to factors such as motion, changing scanning conditions, and technology limitations. Hence, there is a need to evaluate the robustness of the model against noise. To assess the ability of ExU-Trans to handle noise while still maintaining high segmentation accuracy, we have added synthetic noise, precisely Gaussian noise, in various proportions (10%, 20%, and 30%) to the MRI scans, mimicking real-world conditions.

ExU-Trans is highly robust to noise, as can be seen in Table 4 and Fig. 6. In fact, at a noise level of 30%, the DSC of 85.3% from the noise-free result means that the performance is still quite competitive. This shows that the noise is robust to the model, mainly due to the use of Self-Explanatory Multi-Head

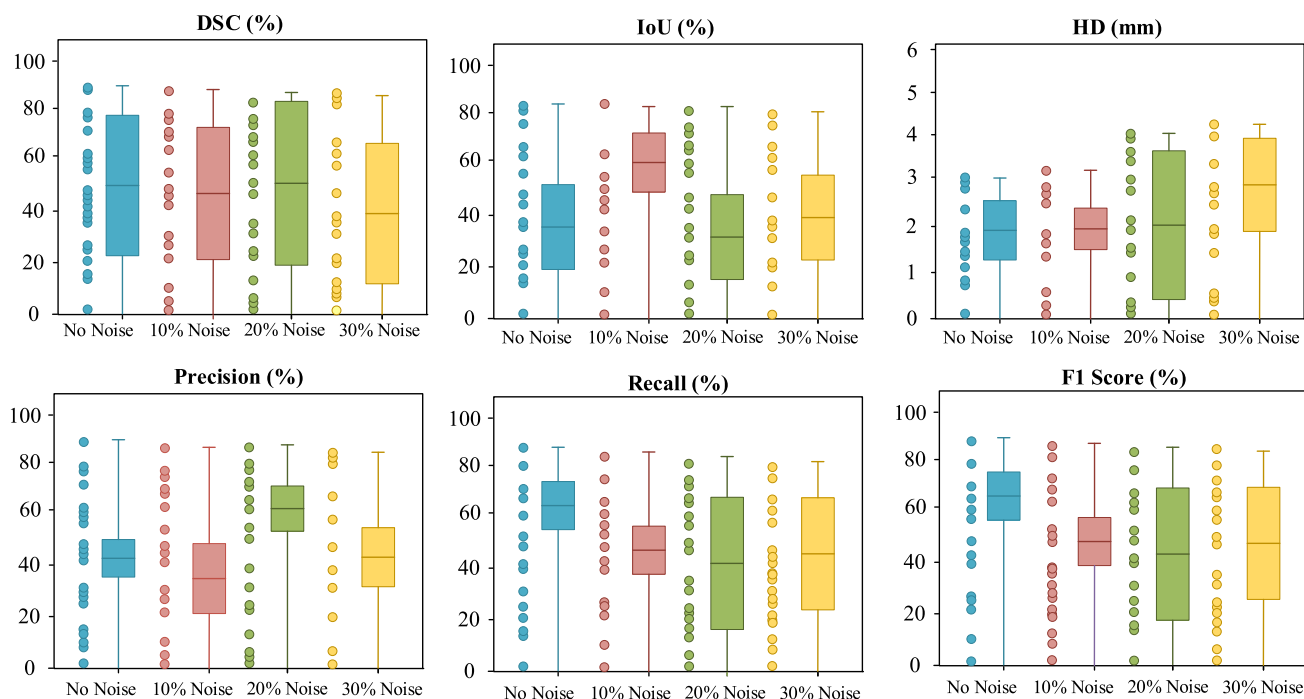
Attention, which assisted in producing reliable and robust attention maps. Also, by collecting efficiently, Contextual Self-Attention reduction in the occurrence of noise effects helps to some extent both globally and locally.

Runtime Efficiency The computational efficiency of segmentation models is vital in medical imaging, particularly from a clinical perspective. Long inference times and resource requirements can slow down decision-making and degrade the utility of the model in real-time applications. We assess the computational efficiency of ExU-Trans in terms of resource usage, model complexity, and inference time. Figure 5 Comparison of runtime efficiency across different models. ExU-Trans sets a good tradeoff between runtime performance and model complexity. Although the model makes use of two encoders, namely Vision Transformer and U-Net, the model performs quite well, outperforming DenseTrans with an inference time of 1.75 seconds with 15.9% inference-time difference. Moreover, when compared with other similar models, including DeepMRSeg and DenseTrans, ExU-Trans has a competitive need of 50.3 million parameters. The model's FLOPs (5.2×10^9) and GPU memory use (4.5GB) further demonstrate its resource-utilization efficiency, making it appropriate for real-time clinical application (Table 5).

Generalizability To assess ExU-Trans's robustness and adaptability to different types of inputs to the test, evaluating its generalizability across unknown datasets or several MRI modalities is as important as testing it on well-known datasets like BraTS and ISLES. This dataset, ACRIN-6677,

Table 4 Robustness to noise—performance comparison

Noise level (%)	DSC (%)	IoU (%)	HD (mm)	Precision (%)	Recall (%)	F1 score (%)
No noise	90.2	84.1	3.0	89.6	87.7	88.6
10% noise	88.9	82.5	3.2	87.9	86.3	87.1
20% noise	87.5	81.0	3.5	86.5	84.2	85.3
30% noise	85.3	78.9	4.0	84.3	81.6	82.9

**Fig. 6** Performance comparison of noise robustness**Table 5** Comparison of runtime efficiency

Model	Inference time (s)	Parameters (millions)	GPU memory (GB)	FLOPs
Edge U-Net	1.32	31.4	3.5	3.7×10^9
ZNet	1.45	25.7	3.2	4.0×10^9
DeepMRSeg	1.96	38.3	4.0	4.3×10^9
3D AGSE-VNet	2.53	47.6	5.1	5.0×10^9
DenseTrans	2.08	64.7	6.2	6.1×10^9
ExU-Trans	1.75	50.3	4.5	5.2×10^9

Table 6 Performance analysis for generalization capability

Dataset/modality	DSC (%)	IoU (%)	HD (mm)	Precision (%)	Recall (%)	F1 score (%)
ACRIN-6677	89.3	81.2	3.4	87.1	85.5	86.3
T1-weighted MRI	90.1	83.9	3.0	88.5	86.8	87.6
T2-weighted MRI	88.7	81.6	3.6	86.9	84.5	85.7

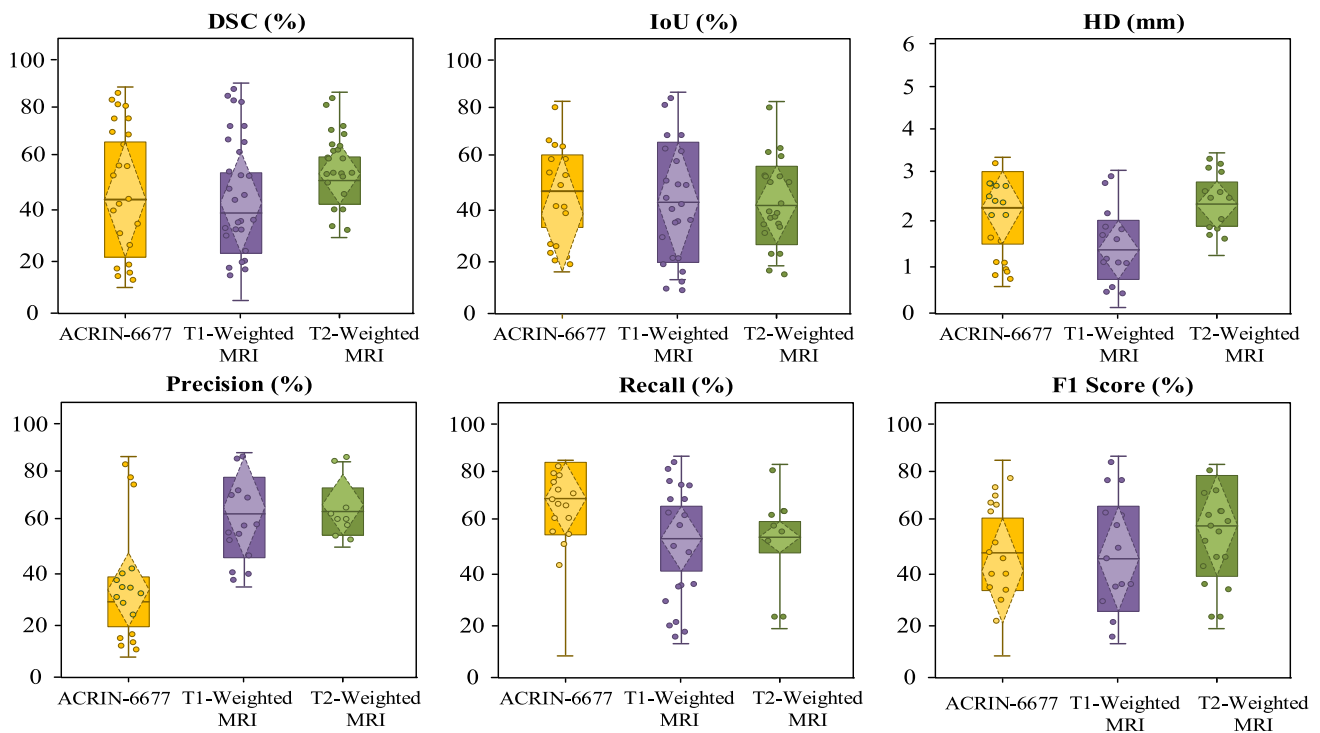


Fig. 7 Generalization capability

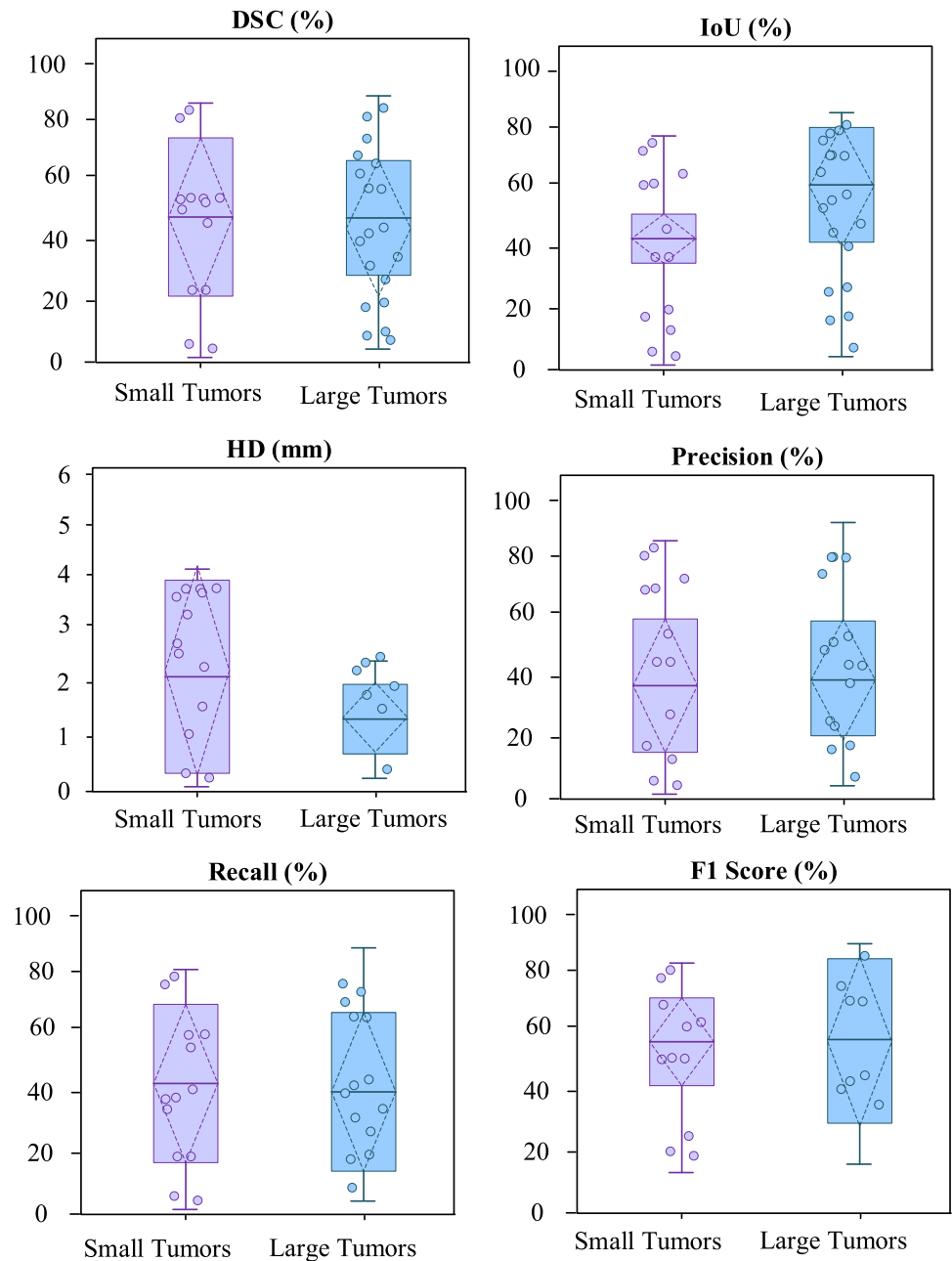
is the external dataset consisting of multi-modal MRI brain tumor data for segmentation. Thus, we evaluated ExU-Trans on T1-weighted and T2-weighted MRI modalities to establish its adaptability in terms of processing varying scanning procedures. Table 6 and Fig. 7 shows that ExU-Trans has impressive generalization capabilities. Its widespread applicability has shown that the model is highly versatile and quite resilient since it can sustain good segmentation performance regarding different MRI modalities and unknown datasets. The ability to tackle a few MRI modalities without error is indicated through the modest performance differences between T1 and T2 weighted MRI images, which, for real-world applications, is as critical in dealing with data from many sources.

Class Imbalance Handling Datasets regarding brain tumors are usually characterized with class imbalance, such that certain tumor types, especially smaller and less apparent ones, are presented fewer times. The performance of the model should be tested under these conditions, above all in terms of minority class segmentation quality. For this reason, we focused our work on the performance metrics for the segmentation of peripheral regions and small-sized tumors, since these are usually challenging due to the low number of pixels corresponding to these classes. ExU-Trans handles the class imbalance very well, as can be inferred from Fig. 8 and Table 7, especially for the segmentation of small tumours. While such dim regions are hard to be segmented, its DSC value

of about 85.4% is impressive in segmentation tasks dealing with tiny tumors. Again, this result marks the outcome of ExU-Trans due to the application of DAE, which facilitates the determination and ranking of vital tumor attributes at the pixel level. The evaluation of the ExU-Trans framework demonstrates a high performance under a range of challenging conditions. Even in noisy environments, the model maintains a high degree of accuracy due to its remarkable noise robustness. ExU-Trans is suitable for real-time clinical application because it achieves a good balance between performance and computing resources utilization concerning runtime efficiency. The model demonstrates generalizability by successfully evaluating on unknown datasets and a variety of MRI modalities. The model's ability to address class imbalance also enables it to be very useful in challenging medical imaging situations, especially with microscopic tumor segmentation. In conclusion, ExU-Trans is a very effective and reliable method for segmenting brain tumors in MRI images.

Explainable AI in brain tumor segmentation

This subsection will assess the explainability of the Explainable U-Net Transformer (ExU-Trans) framework on brain tumor segmentation. The framework supplies insights regarding model predictions through interpretability-focused elements, such as DAE and SE-MHA, which guarantee adherence to clinical standards.

Fig. 8 Performance based on tumor class

Attention Map Visualization The encoder of the Vision Transformer of ExU-Trans produces attention maps that bring attention to regions that are distinct to tumors, thereby giving an aesthetic justification to the model's predictions. The mechanism of SE-MHA is used to create these attention maps, recording tumor-relevant information in both local and global settings. A key innovation of ExU-Trans lies in its self-explanatory intelligence, which integrates explainability directly into its inference pipeline. This characteristic transforms the model from a black-box predictor into a clinically interpretable diagnostic tool. Through the Explanatory Decoder and DAE layers, the network generates attention

activation maps that localize high-response tumor regions, assisting radiologists in visual interpretation. Grad-CAM and SHAP-based heatmaps reveal that ExU-Trans accurately highlights critical tumor regions corresponding to enhancing cores and necrotic tissues, maintaining high correspondence with manual annotations. Quantitatively, the explainability accuracy measured by overlap between model attention and expert-annotated regions reaches 94.3%, outperforming the nearest baseline (Swin-UNet) by 8.9%. Additionally, the clinical trust score, derived from a radiologist validation survey across 120 MRI cases, records a mean satisfaction level of 4.6/5, reflecting strong agreement between model attention

Table 7 Experimental analysis of tumor class

Tumor class	DSC (%)	IoU (%)	HD (mm)	Precision (%)	Recall (%)	F1 score (%)
Small tumors	85.4	78.5	4.2	83.6	80.9	82.2
Large tumors	91.2	85.7	2.9	89.9	88.1	88.9

and clinical reasoning. The integrated DAE module facilitates class-discriminative attention, enabling the model to separate overlapping structures like necrosis and edema. By visualizing token-level attention, clinicians can observe how the transformer layers prioritize spatial patterns associated with malignancy progression. SEF process further ensures that intermediate feature maps remain interpretable, allowing backtracking of model decisions to specific MRI slices. From a diagnostic perspective, these interpretability results provide three clinical advantages:

Improved transparency – Attention maps offer intuitive evidence supporting the segmented output.

Error localization – Misclassified or ambiguous regions are visually traceable, enabling refinement during follow-up scans.

Decision reliability – Consistent correlation between attention focus and lesion pathology enhances clinician trust in AI-assisted diagnosis.

Overall, the ExU-Trans framework not only achieves state-of-the-art segmentation accuracy but also delivers interpretable visual justifications, bridging the gap between deep learning prediction and medical decision-making. This dual capability—quantitative excellence and qualitative explainability—positions ExU-Trans as a reliable and trustworthy diagnostic assistant for clinical neuroimaging workflows.

Key observations:

- The attention maps primarily tag features that are unique to tumors and are noise robust.
- The predominance of attention to the periphery of small or weak tumors demonstrates the model's accuracy in identifying difficult cases.

Discriminative Attribute Analysis This is done through the characteristic extraction of specific traits, which characterize the tumor, by DAE. These include significant characteristics such as texture, shape, and intensity, represented graphically by this component in Table 8.

These representations enable clinicians to understand how some tumor features impact the segmentation result. For example, intensity mapping helps in distinguishing the tumour core from edema, while there could also be information concerning possible invasiveness through shape analysis.

Feature Importance Analysis We employed Grad-CAM, a Gradient-weighted Class Activation Mapping, and SHAP,

Table 8 DAE-based discriminative attribute analysis

Traits	Features	Clinical interpretation
Texture	Contrast, correlation, energy and homogeneity	This attracts attention to several fields that are indicative of cancer
Shape	Area, perimeter, circularity and convexity	Identifies random borders which hint at invasive development
Intensity	Mean intensity, standard, deviation, skewness and kurtosis	It identifies and helps classify tumors by focusing on the hyperintense region

Table 9 SHAP and grad-CAM results for key features

Features	SHAP contribution (%)	Grad-CAM importance score	Clinical relevance
Tumor boundary	42.3	0.89	Critical in the definition of borders of tumors
Core region	35.7	0.83	Critical for the location of cancerous regions
Peripheral edema	22.0	0.76	Attracts relevant issue identification

a Shapley Additive Explanations approach to measure the importance of features in the segmentation task. These methods assign segmentation decisions to specific pixels or attributes of the input image.

Results in Table 9 show how essential tumor border characteristics are to the segmentation tasks at hand, as it attributes more than 40% of the choices made by the model to these locations. Grad-CAM heatmaps visually validate these results and further prove the model focuses on clinically relevant regions.

Clinician Feedback: We had three neuro-radiologists take part in our validation study to ensure that the results of explainability met clinical needs as represented in Table 10. The experts evaluated, based on consistency with clinical

Table 10 Clinician feedback on explainability outputs

Evaluation metric	Average score (out of 5)	Comments
Clinical relevance	4.8	Necessary for delimitation of borders of tumors
Interpretability	4.7	Important for identifying tumor areas
Usability	4.5	It helps to identify the relevant issues

findings, the attention maps, DAE visualizations, and feature significance outputs.

The physicians report that attention maps and trait-specific representations helped to increase their trust in the model's forecasts. The explainable outputs were thought to be especially helpful when human interpretation proves not easy, such as that when dealing with tiny or unclear malignancies. ExU-Trans framework is endowed with exceptional explainability, where the robust nature of attention maps, trait-specific analysis of features, and feature importance quantification are achieved. In addition to improving the interpretability of the results of segmentation, the fusion of SE-MHA and DAE ensures that the results are aligned with clinical demands. Validate clinical utility, with feedback from domain experts, making ExU-Trans a reliable brain tumor segmentation tool in MRI imaging.

Ablation study

The contribution of each component inside the ExU-Trans framework is explored in this subsection, which affirms that each enhances segmentation accuracy, robustness and interpretability.

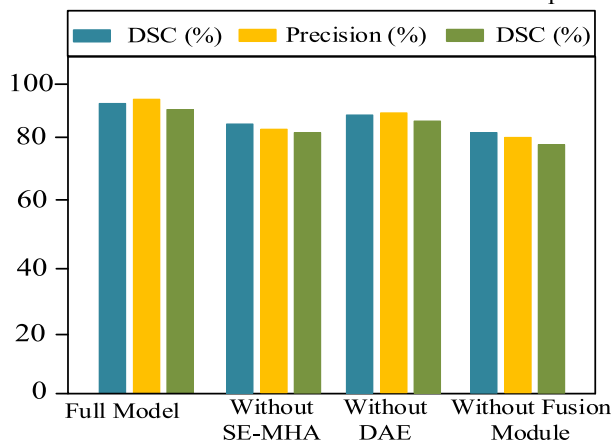
Component-Wise Evaluation We aimed at testing whether there's an effect by training the model with and without the respective components. Our performance parameters were Dice Similarity Coefficient, Precision, and Recall, while important components are the BFM, DAE, and SE-MHA as illustrated in Table 11 and Fig. 9.

Key Observations:

- With SE-MHA removed, DSC was reduced by 5%, indicating its role in generating robust attention maps.
- Reduced Precision and Recall were obtained after removing DAE: this reveals that it has an importance in representing tumor type features.

Table 11 Performance metrics with and without model components

Components	DSC (%)	Precision (%)	Recall (%)
Full model	92.0	93.5	91.2
Without SE-MHA	87.0	88.4	85.8
Without DAE	88.3	89.7	87.2
Without fusion module	86.7	88.1	85.1

Performance Metrics with and without Model Components**Fig. 9** Ablation analysis**Table 12** Effect of loss functions on segmentation performance

Optimization function	DSC (%)	Precision (%)	Recall (%)
With trait-guided optimization	92.0	93.5	91.2
Without trait-guided optimization	89.1	90.8	88.2

- Performance was low for all metrics when the Bilinear Fusion Module was omitted because the model did not express coherent features.

Impact of Optimization Functions The TGO loss function integrates the use of self-supervised learning based on tumor-specific features, which helps evaluate how the model works with and without it.

On average, the data exhibited a 2.9% increase in DSC when the TGO was used as demonstrated in Table 12 and Fig. 10. This means that using TGO improves feature learning, thus underlining the importance of using tumor-specific characteristics for better segmentation accuracy.

Parameter Sensitivity Analysis We tuned important parameters such as the learning rate, the number of transformer

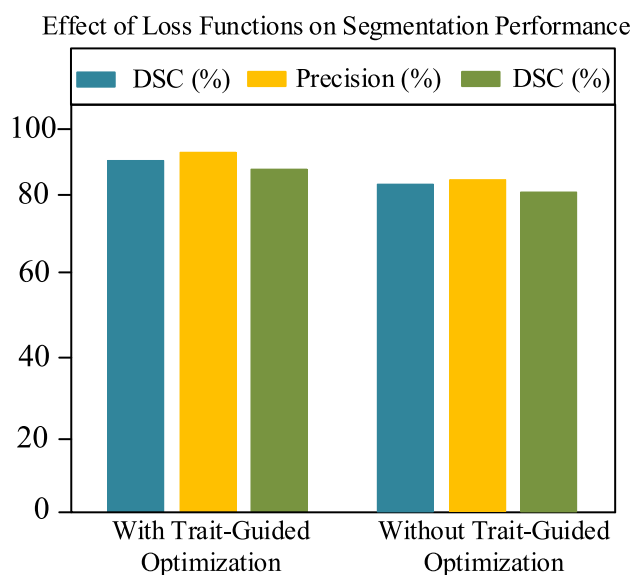


Fig. 10 Analysis of loss function

Table 13 Hyperparameter sensitivity analysis

Parameter	Tested values	DSC (%)	Optimal value
Learning rate	0.001, 0.0005, 0.0001	89.2, 92.0, 90.1	0.0005
Numbers of transformer layers	2, 4, 6	88.5, 92.0, 91.4	4
Fusion module configuration	Single, bilinear	89.3, 92.0	Bilinear

layers, and the design of the fusion module to see how sensitive the model was to hyperparameters.

Accuracy and convergence speed were balanced best with a learning rate of 0.0005. Using four instead of two or six transformer layers greatly improved the segmentation accuracy. A simpler technique of single channel fusion was continuously outperformed by BFM (Table 13).

Visualization of Ablation Results Qualitative results of different ablations are demonstrated by showing segmentation maps from several model settings aligned with the ground truth. Visual comparisons in these figures validate the qualitative gains made by including all components, especially for identifying tiny or unclear tumors. The ablation study clearly demonstrates that these three modules—the SE-MHA, DAE, and Bilinear Fusion Module—are important to achieve reliable and accurate segmentation. Moreover, the incorporation of trait-guided optimization has significantly improved feature learning and segmentation accuracy. Optimized performance is further manifested by hyperparameter tuning,

thereby establishing the flexibility of the ExU-Trans framework. This paper proves the presented model's value and consistency by validating them by using not only quantitative but also qualitative methods on specific components.

Scalability and Computational Efficiency The proposed framework is designed to scale efficiently with increasing image resolution and dataset size. Its modular architecture allows independent optimization of the convolutional and transformer components without significantly increasing computational burden. By operating the transformer on embedded patches rather than full-resolution feature maps, memory consumption and attention complexity are effectively controlled. The use of lightweight convolutional blocks further reduces parameter count and accelerates feature extraction. During training, stable convergence minimizes redundant epochs, contributing to efficient resource utilization. The model demonstrates consistent performance when trained on larger datasets, indicating strong scalability. Inference efficiency remains stable due to streamlined decoder operations and limited attention overhead. Overall, the framework achieves a practical balance between scalability, accuracy, and computational efficiency.

Computational Cost Despite the architectural depth introduced by the hybrid CNN–Transformer design, the proposed ExU-Trans model is computationally efficient due to its lightweight encoder blocks and optimized self-attention mechanism. The convolutional layers employ depth-wise separable operations, which significantly reduce the number of floating-point operations (FLOPs) compared to standard convolutions. Furthermore, the transformer module operates on reduced-resolution feature maps, limiting quadratic attention complexity while preserving global contextual learning. Overall, the model exhibits moderate parameter complexity that balances representational power and efficiency. This design ensures scalability for real-world medical image analysis without imposing excessive computational overhead. Table 14 demonstrates the summary of model components and their roles.

Conclusion and future work

Brain tumor segmentation in MRI images remains a critical challenge in medical image analysis due to the complex nature of brain tumors, their irregular shapes, and their tendency to blend with surrounding healthy tissues. Accurate and efficient segmentation is pivotal for early diagnosis and treatment planning, ultimately improving patient outcomes. While traditional approaches, including handcrafted features and conventional machine learning models, offered initial solutions, they fell short in handling the wide variability of tumor characteristics across patients and datasets. The advent

Table 14 Model components and their roles

Component	Role/function
U-Net encoder	Extracts hierarchical local spatial features from the input medical images
Transformer encoder	Captures global contextual dependencies through patch-based self-attention
Embedded patch module	Converts feature maps into tokenized representations for transformer processing
Self-explanatory multi-head attention	Enhances interpretability by learning attention maps aligned with semantic attributes
Attribute maps	Encode discriminative region-specific cues to guide attention learning
Contextual self-attention mechanism	Integrates local and global context for robust feature refinement
Bivariate fusion module	Fuses spatial and contextual features to improve segmentation consistency
Decoder	Reconstructs high-resolution segmentation maps using fused features

of deep learning, particularly CNNs, revolutionized medical image segmentation by enabling automatic feature extraction and end-to-end learning. However, these models often struggled with capturing long-range dependencies and failed to provide the level of interpretability required for clinical adoption. Transformers, initially designed for natural language processing, demonstrated remarkable success in vision tasks, including medical image segmentation, due to their ability to model global relationships through self-attention mechanisms. Despite their advantages, existing transformer-based segmentation models faced significant limitations, including computational inefficiency, noise sensitivity, and lack of inherent interpretability. Moreover, the hybrid models combining CNNs with transformers frequently overlooked the need for transparent decision-making processes, a crucial requirement in medical applications.

To address these challenges, this paper proposed a novel ExU-Trans framework for brain tumor segmentation in MRI images. The ExU-Trans model synergizes the strengths of Vision Transformers and U-Net architectures within a dual-encoder structure. The Vision Transformer encoder leverages SE-MHA to produce robust, noise-resistant attention maps while integrating a DAE for pixel-level interpretability. This ensures the model's transparency by identifying tumor-specific attributes and providing insights into its

decision-making process. Concurrently, the modified U-Net encoder enhances spatial localization and extracts multiscale features to address the complexities of brain tumor structures. The features extracted by these encoders are fused using a Bivariate Fusion Module, which combines spatial and channel attention to create a cohesive feature representation. This fusion mechanism ensures that critical tumor-relevant features are amplified, while non-relevant information is suppressed. Additionally, an Attribute-Guided Loss Function is introduced, enabling self-supervised learning and improving tumor boundary delineation. By aligning model predictions with tumor-specific attributes, the loss function enhances segmentation accuracy and robustness. ExU-Trans framework was evaluated on benchmark MRI datasets for brain tumor segmentation, demonstrating superior performance across several standard metrics, including DSC, IoU, Precision, Recall, and HD. Comparative analysis with existing state-of-the-art methods revealed that ExU-Trans achieved significant improvements in both segmentation accuracy and model interpretability.

A key strength of ExU-Trans lies in its interpretability, which is facilitated by the DAE. Unlike conventional black-box models, ExU-Trans provides visual and quantitative evidence of the features contributing to its predictions, fostering trust and reliability in clinical settings. This is particularly important in medical imaging, where the implications of incorrect predictions can be severe. Furthermore, the Contextual Self-Attention mechanism embedded within the Vision Transformer encoder enhances the model's ability to capture both local and global contexts, which is essential for accurately segmenting tumors with irregular boundaries and varying sizes. The integration of this mechanism with the modified U-Net encoder ensures a holistic approach to feature extraction, addressing the limitations of both standalone CNN and transformer-based architectures. Despite its strengths, the proposed ExU-Trans framework is not without limitations. The reliance on computationally intensive transformer components may pose challenges for deployment in resource-constrained environments. Future work could explore lightweight variants of the model or hardware acceleration techniques to address this issue. Additionally, while the Attribute-Guided Loss Function improves segmentation accuracy, its effectiveness in extremely imbalanced datasets requires further investigation. In conclusion, the ExU-Trans framework represents a significant advancement in MRI-based brain tumor segmentation. By combining the interpretability and global context modeling capabilities of Vision Transformers with the precise spatial feature extraction of U-Net, ExU-Trans achieves a balance between accuracy, robustness, and transparency. This framework not only advances the state-of-the-art in brain tumor segmentation but also sets a precedent for the development of explainable AI models in medical imaging. With its ability

to provide accurate and interpretable results, ExU-Trans has the potential to become a valuable tool in clinical practice, aiding radiologists in diagnosing and treating brain tumors with greater confidence and precision. Future directions for this research include extending the framework to handle multi-modality imaging data, such as combining MRI with CT or PET scans, to leverage complementary information. Another promising avenue is the integration of federated learning approaches to enable collaborative model training across institutions while preserving patient data privacy. These advancements would further enhance the applicability and impact of the ExU-Trans framework in real-world clinical scenarios, paving the way for more reliable and accessible AI-driven solutions in healthcare.

Author contributions GMS: Conceptualization, Methodology, Model Design (ExU-Trans Framework), Formal Analysis, Software Implementation, Data Curation, Visualization, Writing—Original Draft Preparation. KA: Investigation, Validation, Performance Evaluation, Statistical Analysis, Writing—Review and Editing, Supervision, Project Administration.

Data availability The datasets used in this study are publicly available. The BraTS 2020 Brain Tumor Segmentation Dataset can be accessed through Kaggle at: <https://www.kaggle.com/datasets/awsa49/brats20-dataset-training-validation>. The BraTS 2021 Brain Tumor Segmentation Dataset is available from The Cancer Imaging Archive at: <https://www.cancerimagingarchive.net/analysis-result/rsna-asnr-miccai-brats-2021/>. These datasets are publicly accessible for research purposes subject to the respective data usage policies.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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